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ALEXANDER GRINMAN RIVERA

Computational Engineering Student | TS Clearance

agrinmanriv0537.github.io

PROFESSIONAL SUMMARY

UT Austin junior building software for physical systems: industrial AI and simulation at Suzano, cybersecurity at Booz Allen Hamilton, and vehicle telemetry and robotics. Python, C++, and Java; bilingual English/Spanish.

EDUCATION

B.S. COMPUTATIONAL ENGINEERING The University of Texas at Austin	Aug 2024 - Present
<i>Cockrell School of Engineering</i>	<i>GPA: 3.54</i>
BUSINESS MINOR The University of Texas at Austin	Jun 2025 - Jul 2025
<i>MSI Program McCombs School of Business</i>	

EMPLOYMENT HISTORY

PROCESS CONTROL INTERN Suzano Packaging	Summer 2026
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- Built the Suzano Process Control LLM with five focused capabilities for control-logic lookup, PI data retrieval, process analysis, simulation, and diagnostics.
- Developed APACS and Rockwell conversion and indexing tools for 6,996 control-logic canvases, preserving source identity and validating generated artifacts.
- Built Python workflows for historical-data cleaning, time alignment, model fitting, held-out validation, and scenario simulation, with Modbus/OpenPLC visualization and read-back verification.
- Implemented data-quality and coverage checks for process analysis, plus bounded background execution to keep the operator interface responsive during long-running requests.

CYBERSECURITY INTERN Booz Allen Hamilton	Jun 2024 - Present
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- Developed Python tooling and REST API integrations for cybersecurity and malware-analysis workflows, collaborating with engineers and analysts through GitLab and Scrum.
- Documented technical workflows for team review and continued development in a cleared environment.

SOFTWARE TEAM MEMBER Longhorn Racing: Baja SAE - UT Austin	Jun 2025 - Present
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- Developed PyQt tools to replay vehicle logs and inspect wheel-speed, steering, pedal, GPS, and IMU data for off-road vehicle testing.
- Collaborated across electrical and mechanical teams on sensor interfaces, hardware-software debugging, and telemetry for design decisions.

ENGINEERING LEADERSHIP

LEAD PROGRAMMER, PRESIDENT FRC Robotics Team 5572	Oct 2021 - May 2024
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- Programmed competition robots in C++ and Java, including vision-based target detection, trajectory calculations, PID tuning, and AprilTag localization.

SKILLS

Programming: Python, C++, Java | **Software:** Git, GitLab, REST APIs, Docker, PyQt, Scrum
Engineering: Process control, system identification, time-series analysis, APACS, Rockwell, OpenPLC, Modbus, telemetry, robotics | **Languages:** English, Spanish

CERTIFICATIONS

PCEP - Python Institute (Apr 2024) | **MTA Python** - Microsoft (Apr 2022)

LINKS

[GitHub: agrinmanriv0537](#) | [LinkedIn: Alexander Grinman Rivera](#)